

```
In [ ]: import marimo as mo
import numpy as np
import matplotlib.pyplot as plt
```

Homework 3

Due Wednesday February 18, 2026

Reading:

- Required: Lecture 4-5 Notes
- Optional: Ch. 2 of IPC

1. (IPC: 2.01) Exercise 2.1

As a particular realization of the above example, consider an 8×8 chessboard and a special piece sitting on it. At any time step, the piece will stay still (with probability $1/2$) or move randomly to one of the neighboring positions (with probability $1/2$). Does this process satisfy the detailed balance condition (2.15)? Which positions on the chess board have lower and higher 'energy'? Compute the partition function.

- $x \in \mathcal{X}$ be a position on the board
- $N(x) \subset \mathcal{X}$ be the set of neighboring positions (reachable in one step).
- $d(x) = |N(x)|$ be the degree of position x .
- Stationary distribution $\pi(x)$ over positions x .
- Transition probabilities:

$$P(y|x) = \begin{cases} 1/2 & \text{if } y = x \\ \frac{1}{2d(x)} & \text{if } y \in N(x) \\ 0 & \text{otherwise} \end{cases}$$

- Balance condition: $\pi(x)P(y|x) = \pi(y)P(x|y)$.

$$\pi(x) \frac{1}{2d(x)} = \pi(y) \frac{1}{2d(y)} \implies \frac{\pi(x)}{d(x)} = \frac{\pi(y)}{d(y)}$$

This distribution does exist when $\pi(x) \propto d(x)$ and therefore satisfies the detailed balance equations

Energy is inversely proportional to probability and therefore inverse with $d(x)$. This means corners are highest energy, edges are middle energy, and center squares are the lowest.

- Corners (4 squares): 3 neighbors.
- Edges not corners ($6 \times 4 = 24$ squares): 5 neighbors.
- Interior ($6 \times 6 = 36$ squares): 8 neighbors.

$$Z = 4(3) + 24(5) + 36(8) = 12 + 120 + 288 = 420$$

2. (IPC: 2.02) Exercise 2.2 A two-level system.

This is the simplest non-trivial example: $X = \{1, 2\}$, $E(1) = \epsilon_1$, $E(2) = \epsilon_2$. Without loss of generality, we assume $\epsilon_1 < \epsilon_2$. This example can be used as a mathematical model for many physical systems, such as the spin-1/2 particle discussed above.

Derive the following for the thermodynamic potentials (where $\Delta = \epsilon_2 - \epsilon_1$ is the energy gap):

$$F(\beta) = \epsilon_1 - \frac{1}{\beta} \log(1 + e^{-\beta\Delta})$$

$$U(\beta) = \epsilon_1 + \frac{\Delta e^{-\beta\Delta}}{1 + e^{-\beta\Delta}}$$

$$S(\beta) = \frac{\beta\Delta e^{-\beta\Delta}}{1 + e^{-\beta\Delta}} + \log(1 + e^{-\beta\Delta})$$

For $\epsilon_1 = -1$ and $\epsilon_2 = 1$, these functions are plotted below (see also Fig. 2.1).

```
In [ ]: def F(beta, epsilon1=-1, epsilon2=1):
    """Free energy per spin."""
    delta = epsilon2 - epsilon1
    return -np.log(np.exp(-beta * epsilon1) + np.exp(-beta * epsilon2)) / be

def U(beta, epsilon1=-1, epsilon2=1):
    """Energy per spin."""
    delta = epsilon2 - epsilon1
    return (
        epsilon1 * np.exp(-beta * epsilon1) + epsilon2 * np.exp(-beta * epsilon2)
    ) / (np.exp(-beta * epsilon1) + np.exp(-beta * epsilon2))

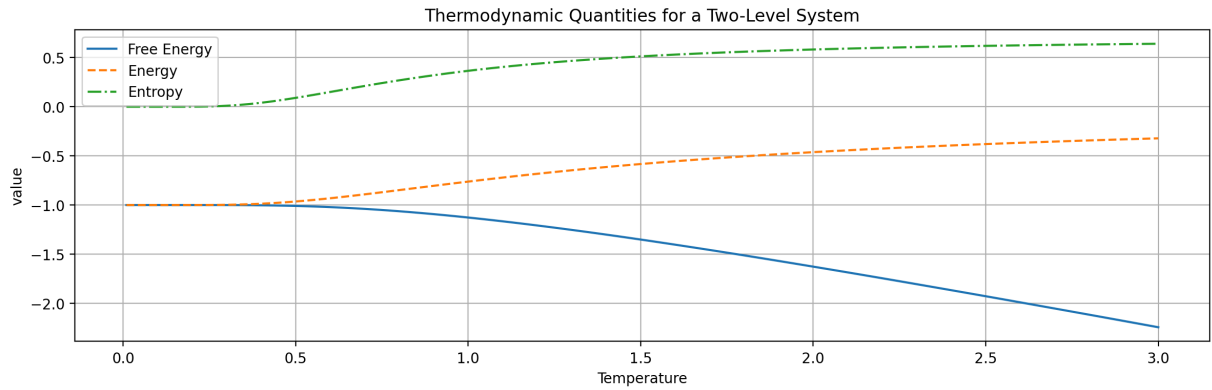
def S(beta, epsilon1=-1, epsilon2=1):
    """Entropy per spin."""
    delta = epsilon2 - epsilon1
    return beta * U(beta, epsilon1, epsilon2) - beta * F(beta, epsilon1, epsilon2)

temperature_range = np.linspace(0.01, 3, 100)
plt.figure(figsize=(12, 4))
plt.plot(temperature_range, F(1 / temperature_range), label="Free Energy")
plt.plot(
    temperature_range, U(1 / temperature_range), label="Energy", linestyle="dashed"
)
plt.plot(
    temperature_range, S(1 / temperature_range), label="Entropy", linestyle="dotted"
```

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)
plt.xlabel("Temperature")
plt.ylabel("value")
plt.title("Thermodynamic Quantities for a Two-Level System")
plt.legend()
plt.grid()
plt.tight_layout()
plt.show()

```



The reader should work out the asymptotics, and verify the general high- and low-temperature behavior shown above.

$$\begin{aligned}
Z(\beta) &= \sum_{x \in \text{set } 1,2} e^{-\beta E(x)} \\
&= e^{-\beta \epsilon_1} + e^{-\beta \epsilon_2} \\
&= e^{-\beta \epsilon_1} (1 + e^{-\beta(\epsilon_2 - \epsilon_1)}) \\
&= e^{-\beta \epsilon_1} (1 + e^{-\beta \Delta}) \\
F(\beta) &= -\frac{1}{\beta} \ln Z(\beta) \\
&= -\frac{1}{\beta} \left(-\beta \epsilon_1 + \ln(1 + e^{-\beta \Delta}) \right) \\
&= \epsilon_1 - \frac{1}{\beta} \ln(1 + e^{-\beta \Delta}) \\
U(\beta) &= -\frac{\partial}{\partial \beta} \ln Z(\beta) \\
&= \frac{\partial}{\partial \beta} [\beta F(\beta)] \\
&= \frac{\partial}{\partial \beta} [\beta \epsilon_1 - \ln(1 + e^{-\beta \Delta})] \\
&= \epsilon_1 - \frac{1}{1 + e^{-\beta \Delta}} \cdot (e^{-\beta \Delta}) \cdot (-\Delta) \\
&= \epsilon_1 + \frac{\Delta e^{-\beta \Delta}}{1 + e^{-\beta \Delta}} \\
S(\beta) &= \beta(U - F) \\
&= \beta \left[\left(\epsilon_1 + \frac{\Delta e^{-\beta \Delta}}{1 + e^{-\beta \Delta}} \right) - \left(\epsilon_1 - \frac{1}{\beta} \ln(1 + e^{-\beta \Delta}) \right) \right] \\
&= \beta \left[\frac{\Delta e^{-\beta \Delta}}{1 + e^{-\beta \Delta}} + \frac{1}{\beta} \ln(1 + e^{-\beta \Delta}) \right] \\
&= \frac{\beta \Delta e^{-\beta \Delta}}{1 + e^{-\beta \Delta}} + \ln(1 + e^{-\beta \Delta})
\end{aligned}$$

3. (FGML: 4.01) Ising Effective field and single-spin conditional.

Consider an Ising model on graph G with energy

$$E(\sigma) = - \sum_{(i,j) \in E} J_{ij} \sigma_i \sigma_j - \sum_i h_i^{\text{ext}} \sigma_i, \quad \sigma \in \{\pm 1\}^n$$

Define $h_i^{\text{eff}} = h_i^{\text{ext}} + \sum_{j \in \partial i} J_{ij} \sigma_j$.

(a) Derive the conditional distribution $P(\sigma_i = +1 | \sigma_{-i})$ in logistic form.

(b) Show that $\mathbb{E}[\sigma_i | \sigma_{-i}] = \tanh(\beta h_i^{\text{eff}})$.

(a) The conditional distribution is given by:

$$\begin{aligned}
 P(\sigma_i | \sigma_{-i}) &= \frac{e^{-\beta E(\sigma)}}{Z(\sigma_{-i})} \propto e^{-\beta E(\sigma)} \\
 E(\sigma) &= -\sigma_i \left(h_i^{\text{ext}} + \sum_{j \in \partial i} J_{ij} \sigma_j \right) \\
 &= -\sigma_i h_i^{\text{eff}} \\
 P(\sigma_i = +1 | \sigma_{-i}) &= \frac{e^{\beta h_i^{\text{eff}}}}{e^{\beta h_i^{\text{eff}}} + e^{-\beta h_i^{\text{eff}}}} \\
 &= \frac{1}{1 + e^{-2\beta h_i^{\text{eff}}}} \\
 &= \sigma(2\beta h_i^{\text{eff}})
 \end{aligned}$$

This is the logistic function (sigmoid) with argument $2\beta h_i^{\text{eff}}$.

(b) The conditional expectation is:

$$\mathbb{E}[\sigma_i | \sigma_{-i}] = (+1)P(\sigma_i = +1 | \sigma_{-i}) + (-1)P(\sigma_i = -1 | \sigma_{-i}) \quad (1)$$

$$= \frac{e^{\beta h_i^{\text{eff}}} - e^{-\beta h_i^{\text{eff}}}}{e^{\beta h_i^{\text{eff}}} + e^{-\beta h_i^{\text{eff}}}} \quad (2)$$

$$= \tanh(\beta h_i^{\text{eff}}) \quad (3)$$

4. (FGML: 4.02) Mean-field self-consistency.

The following fully-connected Ising model on n spins, known as the Curie–Weiss model, is defined by

$$E(\sigma) = -\frac{J}{n} \sum_{1 \leq i < j \leq n} \sigma_i \sigma_j - h \sum_{i=1}^n \sigma_i, \quad \sigma_i \in \{\pm 1\}, J > 0.$$

Let $m = \frac{1}{n} \sum_i \sigma_i$ be the magnetization.

(a) In this model, one assumes that the magnetization concentrates around its average $m \approx \frac{1}{n} \sum_i \mathbb{E}[\sigma_i]$. Then, this is used to compute new means $\mathbb{E}[\sigma_i | \sigma_{-i}]$ based on the other spins. Derive the self-consistency equation for m implied by equilibrium.

(b) For $h = 0$, determine the range of β for which the $m = 0$ fixed point is stable.

(a) The effective field on spin i is:

$$h_i^{\text{eff}} = h + \sum_{j \neq i} \frac{J}{n} \sigma_j$$

Approximating the sum using the mean magnetization $m = \frac{1}{n} \sum_k \sigma_k$:

$$\sum_{j \neq i} \frac{J}{n} \sigma_j \approx \frac{J}{n} (nm) = Jm$$

(ignoring the $O(1/n)$ correction for excluding i).

So $h_i^{\text{eff}} \approx h + Jm$.

From Problem 3, the equilibrium expectation for a single spin in a fixed field is $\tanh(\beta h^{\text{eff}})$. Self-consistency requires the average magnetization to match this expectation:

$$m = \tanh(\beta(h + Jm))$$

(b) For $h = 0$, the equation is $m = \tanh(\beta Jm)$. The fixed point $m = 0$ is always a solution ($\tanh(0) = 0$).

For $m = 0$ to be a stable fixed point, we need the slope to be less than 1 (so the contraction maps m back to 0). The slope of the RHS at $m = 0$ is:

$$\left. \frac{d}{dm} \tanh(\beta Jm) \right|_{m=0} = \text{sech}^2(0) \cdot (\beta J) = \beta J$$

$$\beta J < 1 \implies \beta < 1/J \quad \text{or} \quad T > J$$

If $\beta J > 1$, the slope is > 1 , and $m = 0$ becomes unstable, with new stable solutions appearing at $m \neq 0$ (ferromagnetic phase transition).

5. (IPC: 2.06) Exercise 2.6

Consider the one-dimensional Ising model in zero field, $B = 0$. Show that when $\delta N < i < j < (1 - \delta)N$, the correlation function $\langle \sigma_i \sigma_j \rangle$ is, in the large- N limit,

$$\langle \sigma_i \sigma_j \rangle = e^{-|i-j|/\xi(\beta)}, \quad \text{where } \xi(\beta) = -\frac{1}{\log(\tanh(\beta))}$$

[Hint: You can either use the general transfer matrix formalism or, more simply, use the identity $e^{\beta \sigma_i \sigma_{i+1}} = \cosh(\beta)[1 + \sigma_i \sigma_{i+1} \tanh(\beta)]$.]

$$\begin{aligned} \mathbb{P}(\sigma) &\propto \prod_k e^{\beta \sigma_k \sigma_{k+1}} = \prod_k \cosh(\beta)(1 + \tanh(\beta) \sigma_k \sigma_{k+1}) \\ Z &= 2^N (\cosh \beta)^{N-1} \end{aligned}$$

$$\begin{aligned}
\langle \sigma_i \sigma_j \rangle &= \frac{1}{Z} \sum_{\sigma} \sigma_i \sigma_j \prod_k \cosh(\beta) (1 + \tanh(\beta) \sigma_k \sigma_{k+1}) \\
&= \frac{2^N \cosh(\beta)^{N-1}}{Z} \sum_{\sigma} \sigma_i \sigma_j \prod_k (1 + \tanh(\beta) \sigma_k \sigma_{k+1}) \\
&= \tanh(\beta)^{j-i} \rightarrow \text{all } \sigma \text{ square to 1 \& appear twice} \\
&= \tanh(\beta)^{|i-j|}
\end{aligned}$$

$$\begin{aligned}
\tanh(\beta)^{|i-j|} &= e^{|i-j| \ln(\tanh \beta)} \\
&= e^{-|i-j| [-\ln(\tanh \beta)]}
\end{aligned}$$

$$\frac{1}{\xi(\beta)} = -\ln(\tanh \beta)$$

$$\xi(\beta) = -\frac{1}{\ln(\tanh \beta)}$$

$$\tanh(\beta)^{|i-j|} = e^{|i-j|/\xi(\beta)}$$

Practice Problems (do not hand in):

1. (IPC: 2.03) Exercise 2.3

We return to the example of the previous section: one water molecule, modeled as a point, in a bottle. We consider the case of a cylindrical bottle of base $B \subset \mathbb{R}^2$ (surface area $|B|$) and height d .

Using the energy function $E(x) = wh(x)$ (see (2.13)), where $h(x)$ is the height of x and w is a constant, derive the following explicit expressions for the thermodynamic potentials:

$$F(\beta) = -\frac{1}{\beta} \log(|B|) - \frac{1}{\beta} \log\left(\frac{1 - e^{-\beta wd}}{\beta w}\right)$$

$$U(\beta) = \frac{1}{\beta} - \frac{wd}{e^{\beta wd} - 1}$$

$$S(\beta) = \log(|B|d) + 1 - \frac{\beta wd}{e^{\beta wd} - 1} - \log\left(\frac{\beta wd}{1 - e^{-\beta wd}}\right)$$

Note that the internal-energy formula can be used to compute the average height of the molecule $\langle h(x) \rangle = U(\beta)/w$. This is a consequence of the definition of the energy (see eqn (2.13) and eqn (2.21)). If we plug in the correct constant w , we can find that the average height falls below 49.99% of the height of the bottle $d = 20\text{cm}$ only when the temperature is below 3.2 K.

2. (IPC: 2.04) Exercise 2.4

Using eqns (2.34)-(2.36), derive the low-temperature expansions:

$$F(\beta) = -\frac{1}{\beta} \log \frac{|B|}{\beta w} + \Theta(e^{-\beta w d})$$

$$U(\beta) = \frac{1}{\beta} + \Theta(e^{-\beta w d})$$

$$S(\beta) = \log \frac{|B|e}{\beta w} + \Theta(e^{-\beta w d})$$

In this case X is continuous, and the energy has no gap. Nevertheless, these results can be understood as follows: at low temperature, the molecule is confined to a layer of height of order $1/(\beta w)$ above the bottom of the bottle. It therefore occupies a volume of size $|B|/(\beta w)$. Its entropy is approximately given by the logarithm of such a volume.

3. (IPC: 2.05) Exercise 2.5

Let us reconsider the above example and assume the bottle to have a different shape, for instance a sphere of radius R . In this case it is difficult to compute explicit expressions for the thermodynamic potentials, but one can easily compute the low-temperature expansions. For the entropy, one gets at large β :

$$S(\beta) = \log \left(\frac{2\pi e^2 R}{\beta^2 w} \right) + \Theta \left(\frac{1}{\beta} \right)$$

The reader should try to understand the difference between this result and eqn (2.39) and provide an intuitive explanation, as in the previous example. Physicists say that the low-temperature thermodynamic potentials reveal the 'low-energy structure' of the system.